

# Workflow Example 3

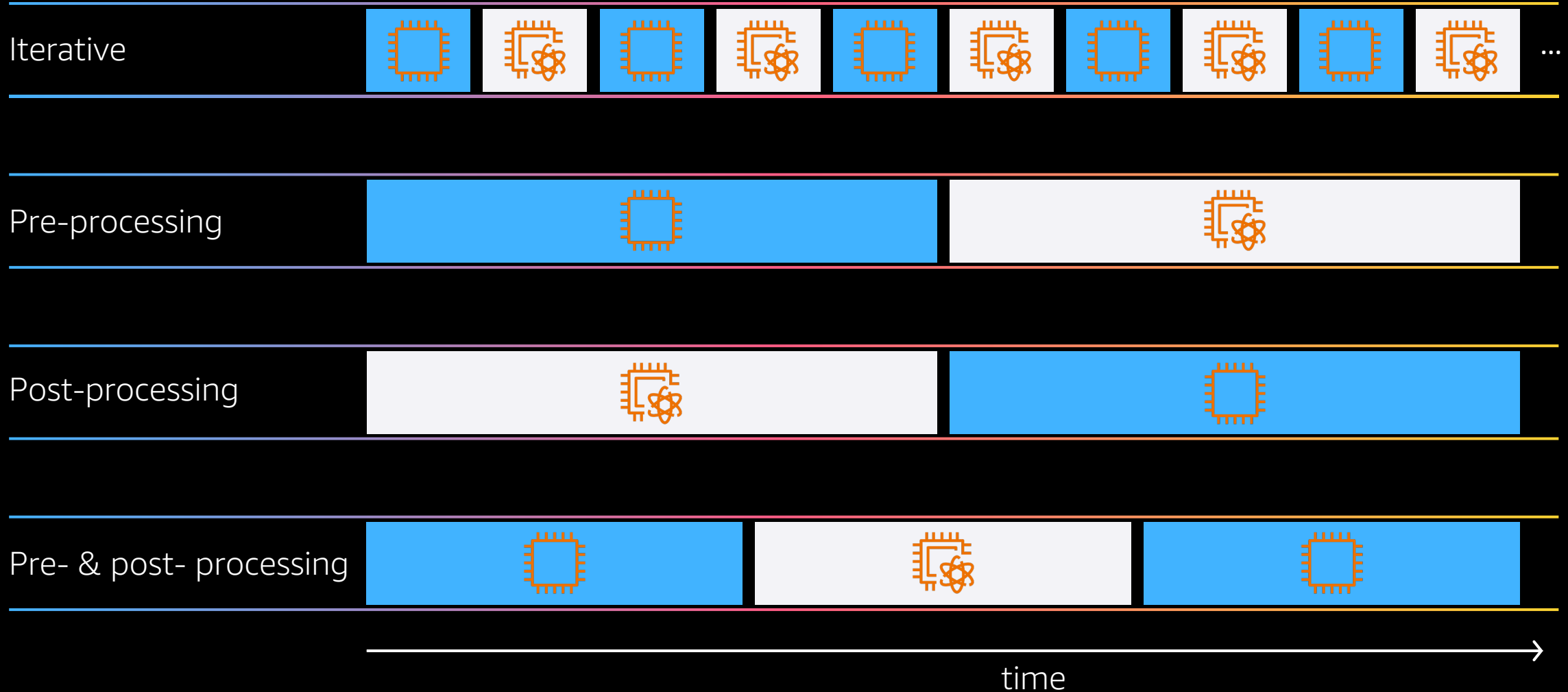
## Generative Quantum Eigensolver

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Worldwide Specialist Organization – Advanced Compute



# Common quantum-classical workflows



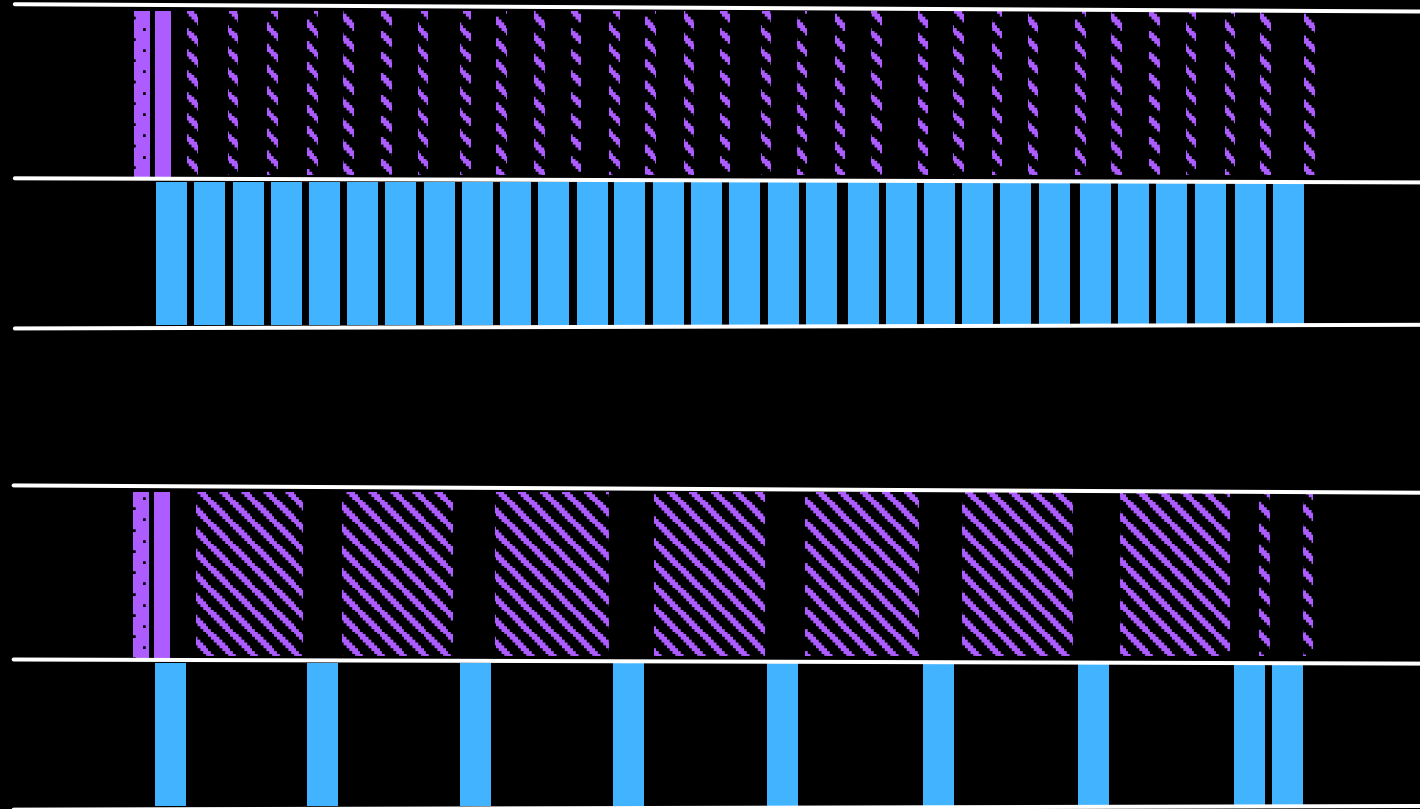
# Iterative quantum-classical workflows

Classical compute

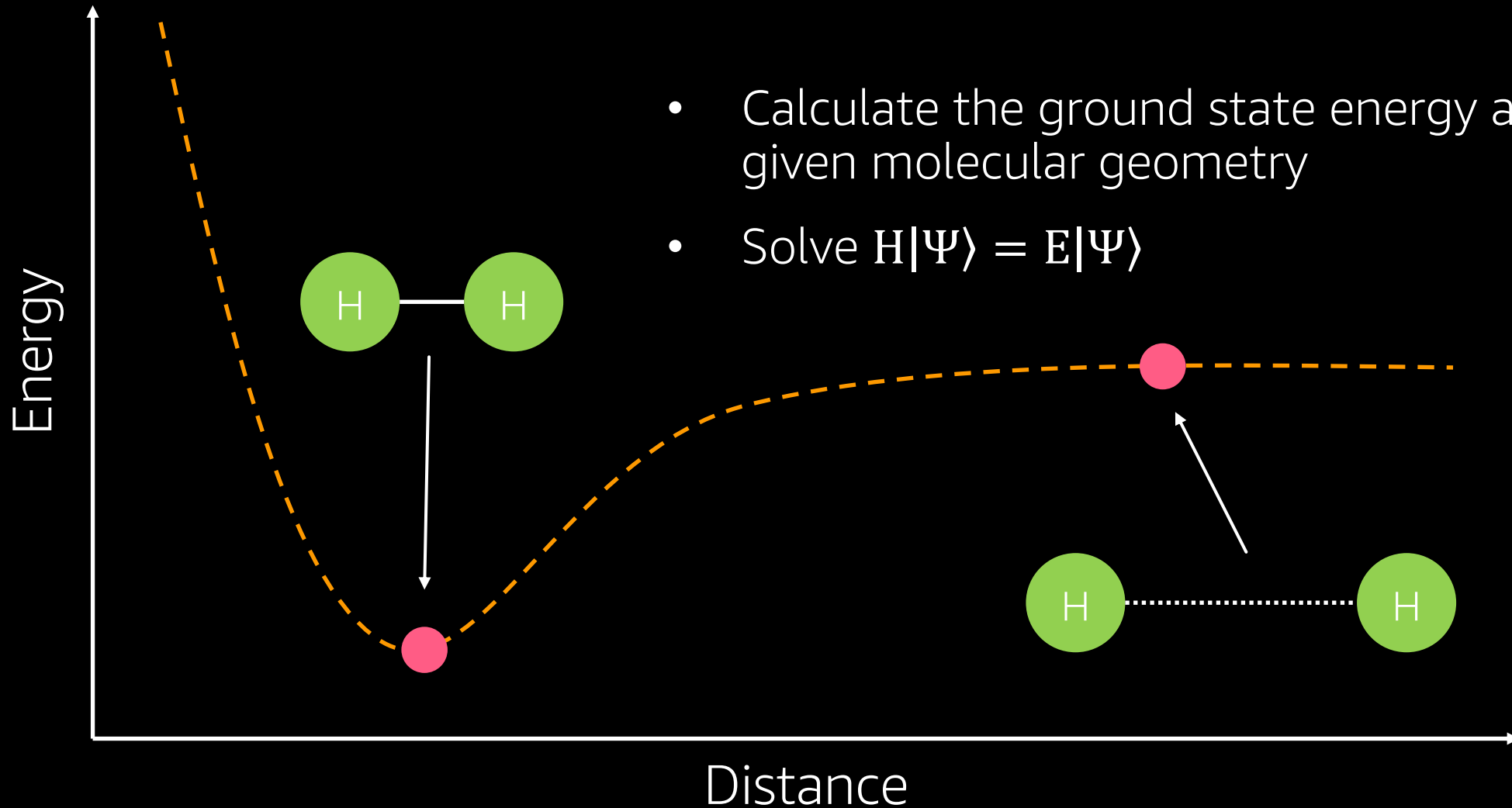
Quantum compute

Classical compute

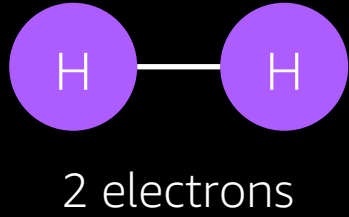
Quantum compute



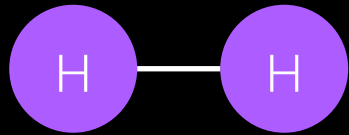
# Quantum chemistry - electronic structure



# Electronic configurations

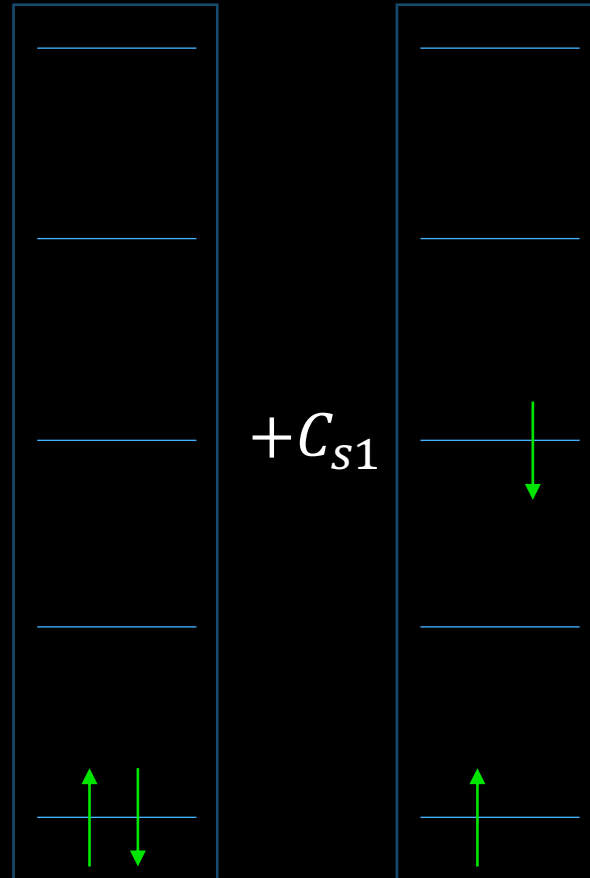


# Electronic configurations

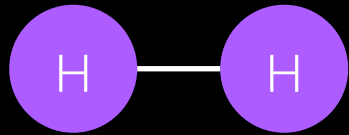


2 electrons

$$|\psi_{approx}\rangle = C_0$$

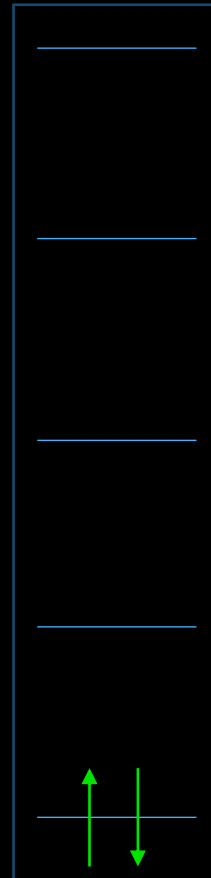


# Electronic configurations

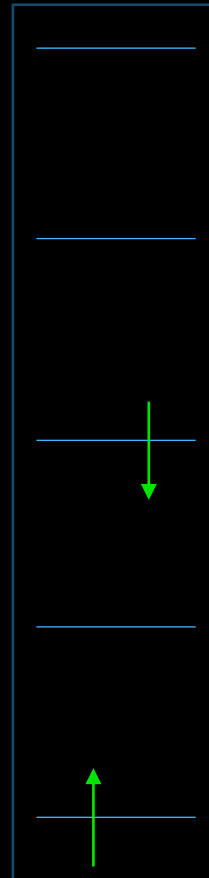


2 electrons

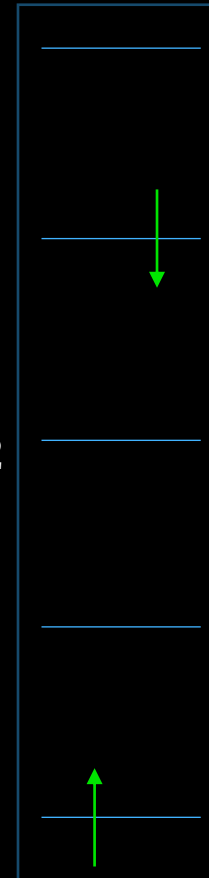
$$|\psi_{approx}\rangle = C_0$$



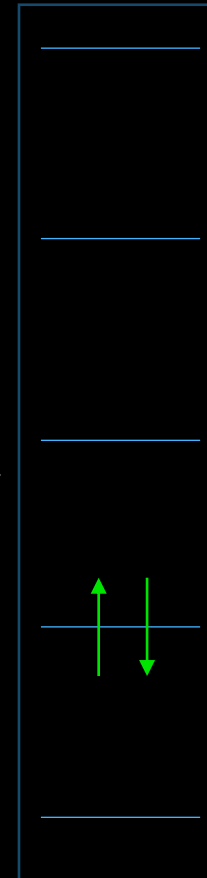
$$+C_{s1}$$



$$+C_{s2}$$



$$\dots + C_{d1}$$



$$+ \dots$$

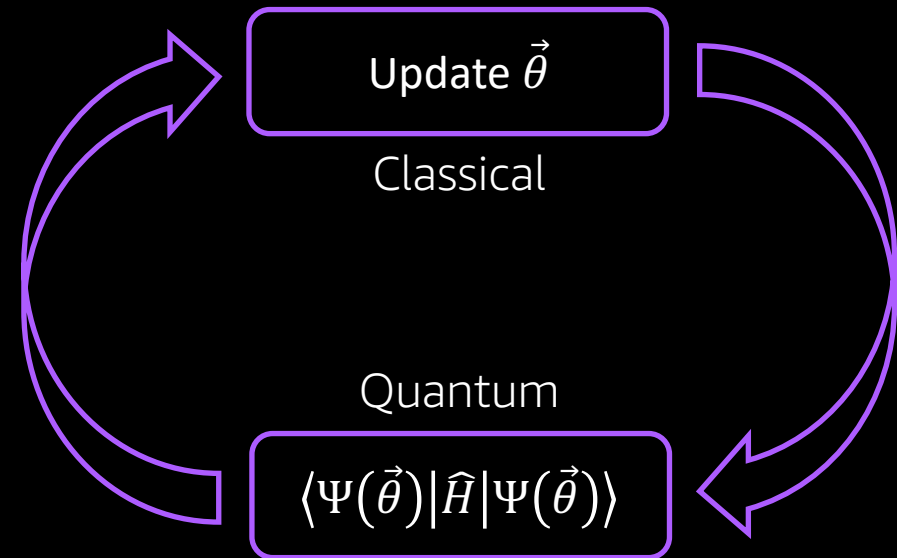
# Variational Quantum Eigensolver - UCCSD

$$|\Psi(\theta)\rangle = e^{\hat{T}(\vec{\theta}) - \hat{T}^\dagger(\vec{\theta})} |\Phi_0\rangle$$

$$\hat{T}(\vec{\theta}) = \hat{T}_{singles}(\vec{\theta}) + \hat{T}_{doubles}(\vec{\theta})$$

$$\hat{T}_{singles}(\vec{\theta}) = \sum_{i,m} \theta_i^m \hat{a}_m^\dagger \hat{a}_i$$

$$\hat{T}_{doubles}(\vec{\theta}) = \sum_{i,j,m,n} \theta_{i,j}^{m,n} \hat{a}_n^\dagger \hat{a}_m^\dagger \hat{a}_j \hat{a}_i$$





# Generative Quantum Eigensolver (GQE) - UCCSD

The GQE algorithm optimizes a classical generative model to produce quantum circuits (from an excitation operator pool) with desired properties (ground state energy).

$$|\Psi(\theta)\rangle = e^{\hat{T}(\vec{\theta}) - \hat{T}^\dagger(\vec{\theta})} |\Phi_0\rangle$$

$$\hat{T}(\vec{\theta}) = \hat{T}_{singles}(\vec{\theta}) + \hat{T}_{doubles}(\vec{\theta})$$

$$\hat{T}_{singles}(\vec{\theta}) = \sum_{i,m} \theta_i^m \hat{a}_m^\dagger \hat{a}_i$$

$$\hat{T}_{doubles}(\vec{\theta}) = \sum_{i,j,m,n} \theta_{i,j}^{m,n} \hat{a}_n^\dagger \hat{a}_m^\dagger \hat{a}_j \hat{a}_i$$

$$|\Psi(\theta)\rangle = e^{\hat{T}(\vec{\theta}) - \hat{T}^\dagger(\vec{\theta})} |\Phi_0\rangle$$

$$\hat{T}(\vec{\theta}) \text{ proposed by generative model}$$

# Generative Quantum Eigensolver - UCCSD

$$\hat{T}_{singles}(\vec{\theta}) = \sum_{i,m} \theta_i^m \hat{a}_m^\dagger \hat{a}_i$$

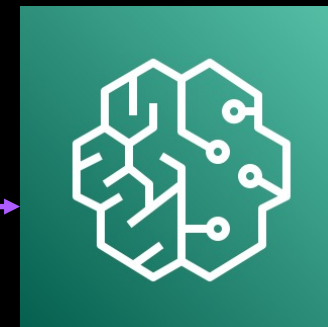
$$\hat{T}_{doubles}(\vec{\theta}) = \sum_{i,j,m,n} \theta_{i,j}^{m,n} \hat{a}_n^\dagger \hat{a}_m^\dagger \hat{a}_j \hat{a}_i$$

$$\Theta = \{\theta_1, \theta_2, \dots, \theta_N\}$$

$$\hat{T}_{singles\ pool} = \{\theta_i^m \hat{a}_m^\dagger \hat{a}_i \mid \theta_i^m \in \Theta\}$$

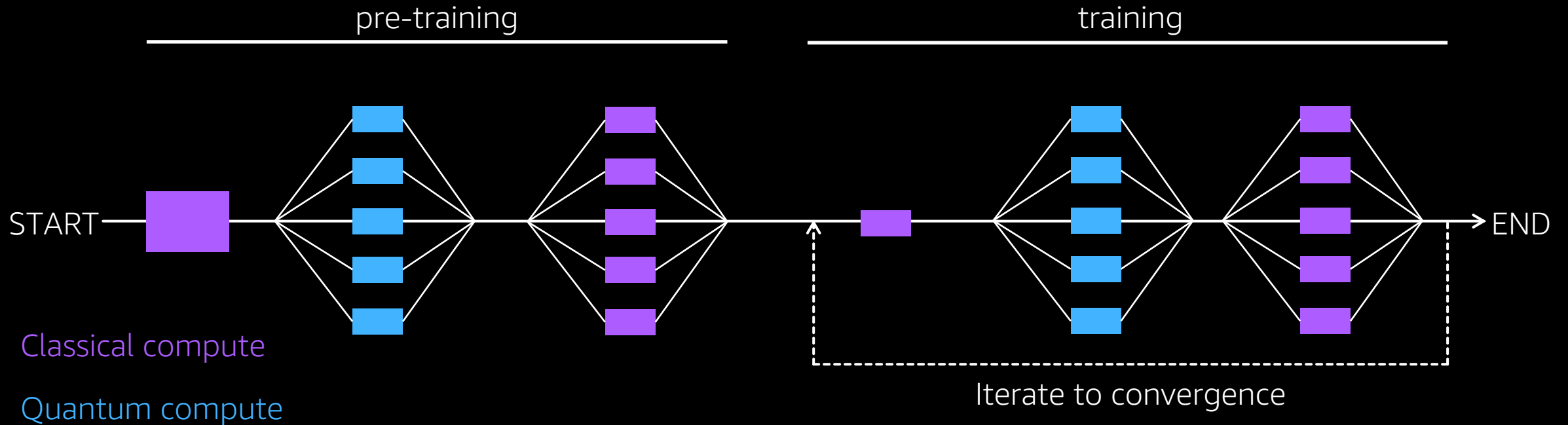
$$\hat{T}_{doubles\ pool} = \{\theta_{i,j}^{m,n} \hat{a}_n^\dagger \hat{a}_m^\dagger \hat{a}_j \hat{a}_i \mid \theta_{i,j}^{m,n} \in \Theta\}$$

$$|\Psi_{GQE}\rangle = e^{\hat{T}_{GQE} - \hat{T}_{GQE}^\dagger} |\Phi_0\rangle$$



model

# GQE model training workflow



# Agree or disagree? Let's discuss

- 1.No hybrid workflows we known today actually need HPC
- 2.No end-user application requires tight integration
- 3.For quantum-classical systems, tight integration does not require co-location